

Decision Trees

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can fit a root question and leaf predictions from training examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

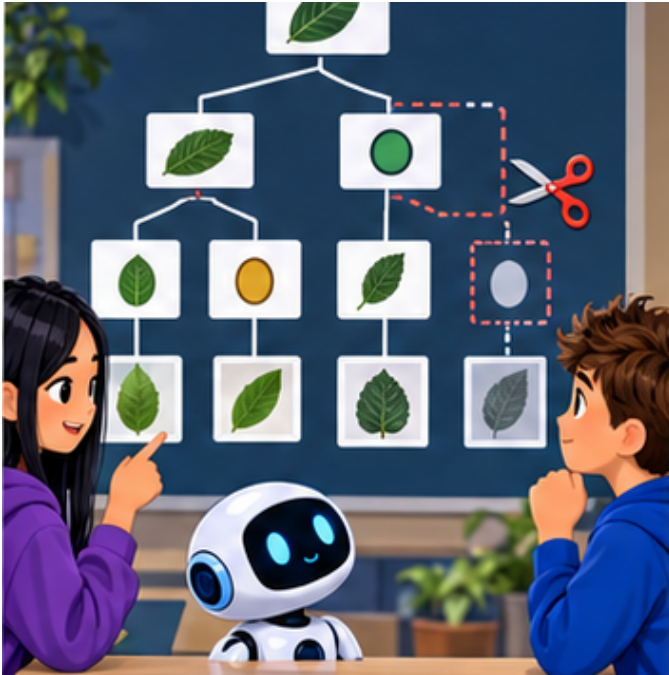
Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A decision tree partitions feature space using ordered questions; each leaf produces a prediction.



LEARNING TARGET Rule: if spots $\geq$ 4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that training found the threshold?

Builder name and experiment date:

Readiness for the connected explanation

Why does a cutoff with zero training errors still need validation on separate examples?

This week's end goal

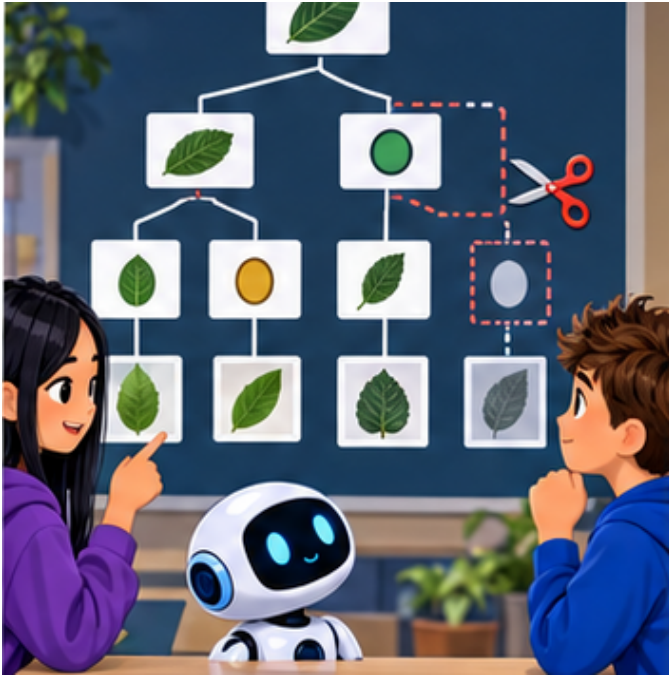
By the end, I can explain decision trees using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use decision trees to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A decision tree asks a feature question and follows a branch to a prediction. A tree with one question has depth one and is sometimes called a stump. Unlike Week 8, where the two output labels were fixed in advance, this week we fit both the question and the label stored at each leaf.

Our six fictional training rows have two binary sensor flags available before prediction: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate. Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row · Near · Tall · Label

Row: 1

Near: 1

Tall: 0

Label: blocked

Row: 2

Near: 1

Tall: 1

Label: blocked

Row: 3

Near: 1

Tall: 0

Label: blocked

Row: 4

Near: 0

Tall: 1

Label: clear

Row: 5

Near: 0

Tall: 0

Label: clear

Row: 6

Near: 0

Tall: 1

Label: blocked

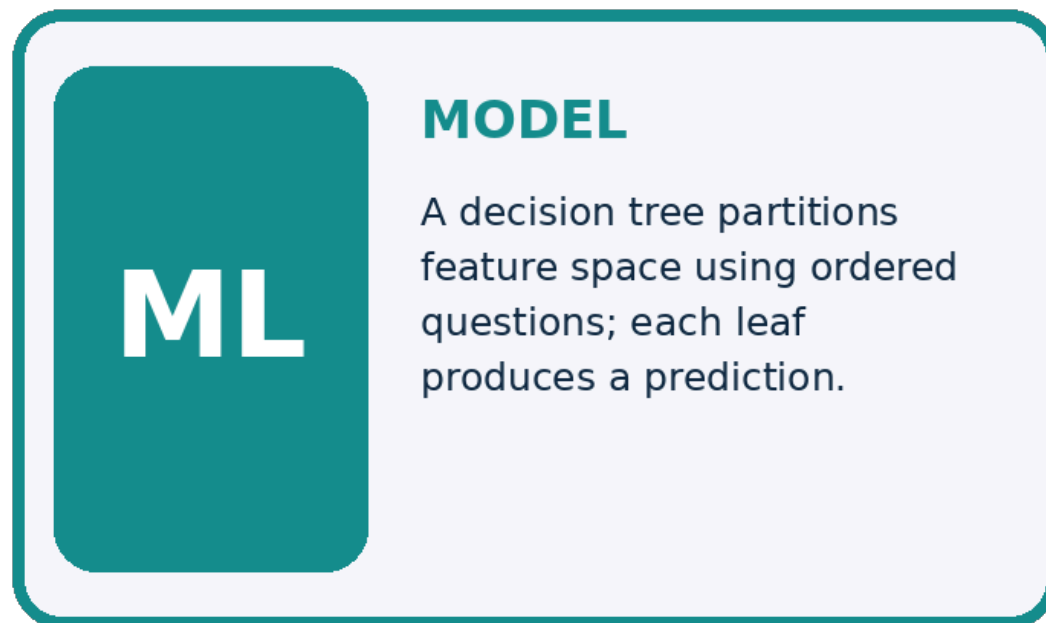
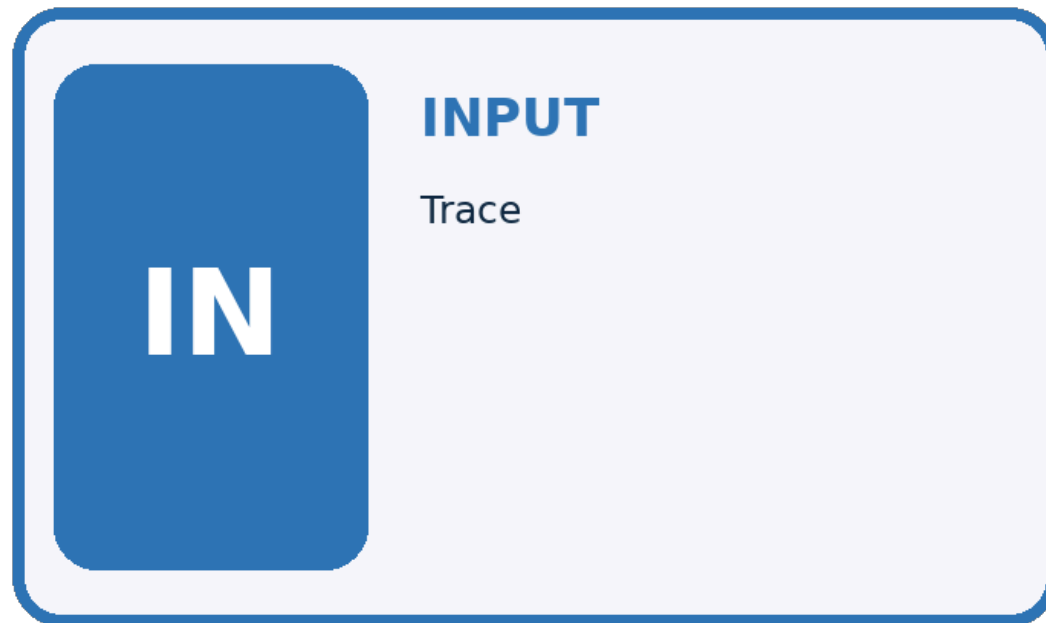
Try near first and tall second. For each question, group the training rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts. Count all training mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom fitting procedure.

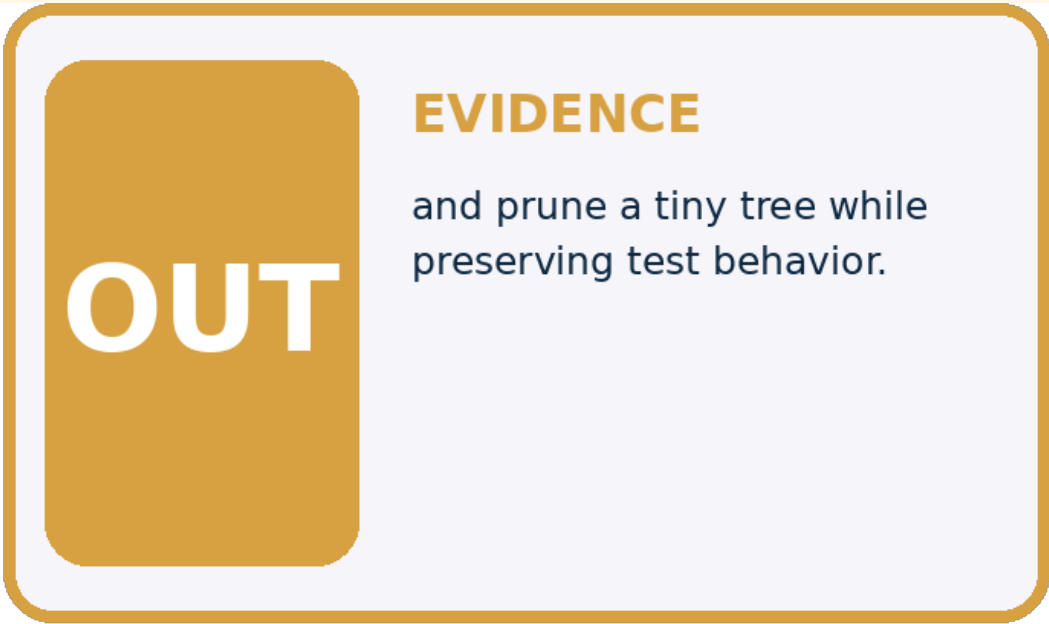
3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

Trace the information, fitted behavior, and evaluation proof separately.





OUT

EVIDENCE

and prune a tiny tree while preserving test behavior.

QUESTION Trace, draw, and prune a tiny tree while preserving test behavior.

MODEL IDEA A decision tree partitions feature space using ordered questions; each leaf produces a prediction.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Root asks whether $\text{spot_count} \geq 3$. Yes branch predicts spotted; no branch asks whether $\text{yellow_ratio} \geq 0.25$.

Draw a fresh data $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ evaluation trace:

A closer explanation

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]

def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"

best_errors = len(rows) + 1

for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
```

```

labels = [row[2] for row in rows

        if row[feature] == value]

leaves[value] = majority(labels)

errors = 0

for row in rows:

    if leaves[row[feature]] != row[2]:

        errors = errors + 1

print(feature, errors)

if errors < best_errors:

    best_errors = errors

    tree = (feature, leaves)

print(tree[0], tree[1][0], tree[1][1])

```

Feature 0 is near; feature 1 is tall; row[2] is the label. Output is 0 1, then 1 2, then 0 clear blocked. Near makes one training mistake: near=0 contains two clear labels and one blocked, so that leaf predicts clear. Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both branches of each candidate. An empty branch would need a declared fallback; this small example does not implement one. The score is total training mistakes, not an average giving small and large branches equal weight.

This is an explicitly limited depth-one learner. General tree algorithms can repeat split selection within branches and often use impurity criteria such as Gini. Our code does not implement that full procedure. Keeping the scope small lets you inspect every decision.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: The stored rule changes to `visible_spots>2`.

Prediction: M is now predicted healthy.

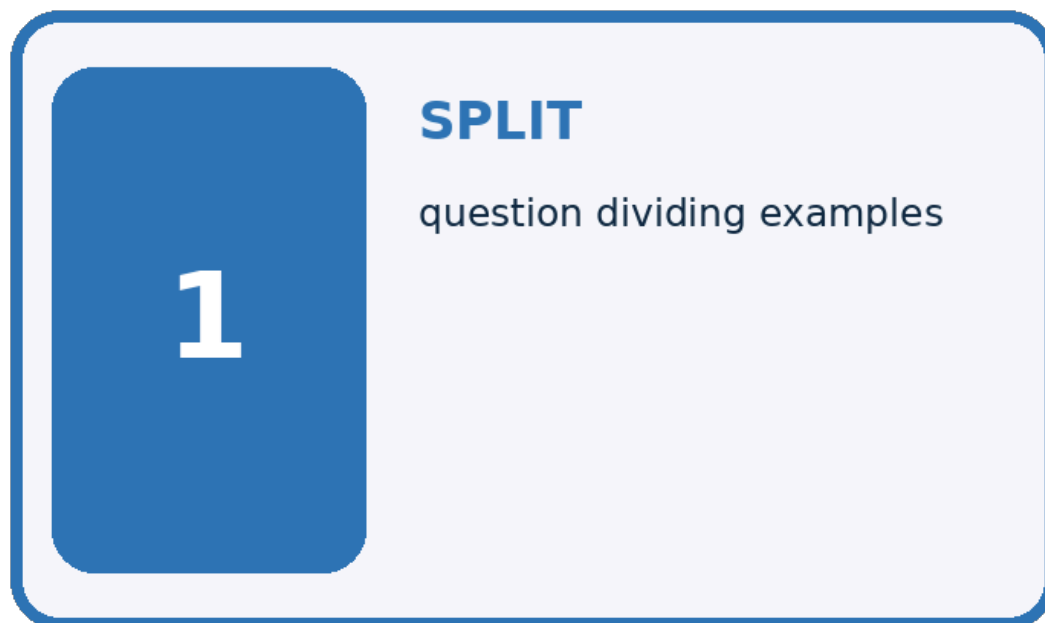
Reason: Changing `>=` to `>` moves the exact value 2 to the other branch.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

THRESHOLD

numeric boundary used in a split

3

LEAF NODE

terminal prediction

4

PRUNING

remove unnecessary
branches

Term · Precise meaning · My ML example

Term: split

Precise meaning: question dividing examples

My ML example: _____

Term: threshold

Precise meaning: numeric boundary used in a split

My ML example: _____

Term: leaf node

Precise meaning: terminal prediction

My ML example: _____

Term: pruning

Precise meaning: remove unnecessary branches

My ML example: _____

MISCONCEPTION TO REPAIR A deeper tree must be better because it asks more detailed questions.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Root

Meaning: The first question asked by a tree

Word: Branch

Meaning: A path selected by the answer to a question

Word: Leaf

Meaning: An endpoint that stores a prediction

Word: Depth

Meaning: The largest number of question steps from root to leaf

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 11 • WORKED EVIDENCE

- 1 Root asks whether $\text{spot_count} \geq 3$.
- 2 Yes branch predicts spotted; no branch asks whether $\text{yellow_ratio} \geq 0.25$.
- 3 Trace leaf A: $\text{spot_count}=1$, $\text{yellow_ratio}=0.10 \rightarrow$ healthy.
- 4 A branch fitting one training oddity hurts test results, so prune and retest.

STEP 1 Root asks whether $\text{spot_count} \geq 3$.

STEP 2 Yes branch predicts spotted; no branch asks whether $\text{yellow_ratio} \geq 0.25$.

STEP 3 Trace leaf A: $\text{spot_count}=1$, $\text{yellow_ratio}=0.10 \rightarrow$ healthy.

STEP 4 A branch fitting one training oddity hurts test results, so prune and retest.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Two paths through the learned tree

Root question · Branch answer · Learned leaf

Root question: Is near equal to 1

Branch answer: No near is 0

Learned leaf: Predict clear

Root question: Is near equal to 1

Branch answer: Yes near is 1

Learned leaf: Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: predictions are blocked, clear, clear, blocked. Two of four match. The training-majority baseline predicts blocked, since training has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted root and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct training predictions out of six is not the validation result.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decision Trees

Trace a small fitted tree, including its exact boundary.

The facts we will use

Toy stored tree: if visible_spots $\geq$ 2 predict spotted; otherwise healthy. Missing visible_spots requires review. Inputs L=1, M=2, N=None.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read first branch

L: $1 \geq 2$ is false, so predict healthy.



2. Read boundary

M: $2 \geq 2$ is true, so predict spotted.



3. Stop missing path

N lacks a count, so use review rather than inventing zero.

CHANGE ONE THING

The stored rule changes to `visible_spots > 2`.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read first branch

What happens and why: L: $1 \geq 2$ is false, so predict healthy.

Step: 2. Read boundary

What happens and why: M: $2 \geq 2$ is true, so predict spotted.

Step: 3. Stop missing path

What happens and why: N lacks a count, so use review rather than inventing zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Applying a stored tree is inference. Selecting a threshold from labelled training data is a separate fitting step.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → **Diagnose** → **Repair**

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: A deeper tree must be better because it asks more detailed questions. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · **Expected** · **Actual** · **Likely cause / next test**

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE A deeper tree must be better because it asks more detailed questions.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Record the two candidate questions, branch label counts, total errors, tie rules, depth limit, stored tree and validation predictions. Retain all rows, including the blocked example in the $\text{near}=0$ branch; removing a disagreement would change the evidence.

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset. Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

A learner says the tree is reliable because $\text{near}=1$ has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Do not assume near=1 must always predict blocked. If a new training set gives that branch a clear majority, this same learner will store clear there. A feature name is not the answer. Use the training labels belonging to that branch.

Adding more questions can fit finer training details but does not guarantee better new-case performance. Record any depth choice made during development, and keep final test data separate. A leaf majority describes this training group; it is not a promise about every future example.

Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example “Fitting a decision tree”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Human Tree Builder: choose splits, trace six examples, compare train/test errors, then prune one weak branch.

1. Write the hypothesis and expected evidence before running.
2. Human Tree Builder: choose splits, trace six examples, compare train/test errors, then prune one weak branch.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

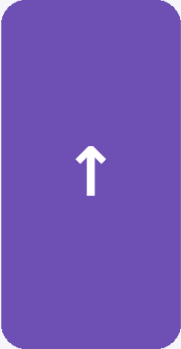
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: split, threshold, leaf node, pruning.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A deeper tree must be better because it asks more detailed questions.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Draw the current Plant Classifier tree and annotate feature, threshold, prediction, and test evidence at each leaf.

THIS WEEK'S CHECKPOINT Draw the current Plant Classifier tree and annotate feature, threshold, prediction, and test evidence at each leaf.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Trace, draw, and prune a tiny tree while preserving test behavior.

REQUIRED PROOF Trace, draw, and prune a tiny tree while preserving test behavior.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Nearest Neighbors.

13. Study a complete case

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A supplied tree predicts spotted for $\text{spot_count} \geq 3$, otherwise healthy.

The question

Trace 2,3,4 and state boundary.

Worked explanation

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned threshold.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace the inclusive tree threshold

Evidence or stage · Value or result

Evidence or stage: 2 spots

Value or result: Below 3 → healthy

Evidence or stage: 3 spots

Value or result: At least 3 → spotted

Evidence or stage: 4 spots

Value or result: At least 3 → spotted

A supplied tree predicts spotted for $\text{spot_count} \geq 3$, otherwise healthy.

Trace 2,3,4 and state boundary.

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned threshold.

14. Try a new case independently

Independent case: Changed independent case

A stored fitted rule says spots ≥ 5 predicts spotted; otherwise healthy. Trace 4,5,6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tree predicts fern when $\text{leaf_width_cm} \leq 3$ and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Draw the current Plant Classifier tree and annotate feature, threshold, prediction, and test evidence at each leaf.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

Transfer practice from the original programme

Rule: if spots ≥ 4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that training found the threshold?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.